Thresholds, coherent structures, and dynamics of critical focusing dispersive systems

Persistent kinetic amplification, conserved occupations, and momentum-adapted motion

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8 October 2026

Abstract

We connect sharp critical concentration geometry to nonlinear evolution through three results. For real-even positive elliptic homogeneous dispersion of order $m > 2$, global zero-momentum negative-energy solutions near critical mass satisfy a lower physical-time mean kinetic bound $\liminf_{T\to\infty} T^{-1}\int_0^T A(t)\,dt \geq c|E|d^{-1-m/[2(m-1)]}$. The result covers critical local powers and a class of strictly positive-definite homogeneous Hartree interactions, without radiality, a spatial moment, a global kinetic bound, or symbol convexity. Fourier factorization preserves an exterior negative kinetic term, while intrinsic-time center construction and weak energy splitting control the small mass tail. For local multicomponent systems, Gaussian projection of the scalar sharp inequality gives an explicit coefficient depending on the entire conserved occupation spectrum. This yields all-time energy bounds for specified above-critical masses, including $M < \sqrt{2}M_c$ for two equal orthogonal occupations in the two-dimensional quartic system, and an exact scalar reduction for every unbounded critical-mass evolution. Finally, near a quartic ground wave with nondegenerate real linearization, momentum-adapted modulation improves the mean bound to $c_Q|E|d^{-7/3}$ at zero momentum and to $c_Q|E|d^{-2}$ at arbitrary momentum. Complete supporting proofs and the remaining dynamical obstruction are included.

The stronger modulation results retain the explicit assumption $\ker L_+ = \operatorname{span}\{\partial_j Q\}$. The general mean-bound theorem states its conservative strong solution class; a concrete local-flow proof is supplied for local powers of order $m = 2k \geq 4$ in dimensions one through three. None of the results proves that all nonradial negative-energy solutions blow up in finite time.

The mathematical connection

The main question is how critical threshold geometry restricts a time-dependent solution. Write $M = \|u\|_2^2$ and $A = \|a(D)^{1/2}u\|_2^2$. Sharp interpolation at critical mass M_c gives only $A(t) \geq 2|E|/d$ when $E < 0$. We show that nonlinear evolution forces a substantially higher *long-time average*. The mass surplus leaves at most $O(d)$ mass outside a persistent core. Quantitative phase control bounds the motion of that core, and the favorable exterior kinetic contribution absorbs the tail current. This produces

$$\liminf_{T\to\infty} \frac{1}{T}\int_0^T A(t)\,dt \geq c|E|d^{-\alpha_m}, \qquad \alpha_m = 1 + \frac{m}{2(m-1)}.$$

For amplitude perturbations $u_d(0) = (1+d)^{1/[2(\kappa-1)]}Q$, where $M(Q) = M_c$ and $\kappa = 1 + m/N$ locally or $\kappa = 2$ for Hartree, this forces mean amplification at least $cd^{-m/[2(m-1)]}$. Every prescribed finite factor is crossed at arbitrarily late times by a global solution for sufficiently small d.

The multicomponent result uses the same threshold geometry in a different dynamical direction. The internal correlation operator $\Gamma = \int |U(x)\rangle\langle U(x)|\,dx$ is conserved. An explicit occupation-sensitive sharp-scalar inequality turns that conservation law into a uniform bound for all times. It

also makes the assertion of scalar behavior at threshold exact for the entire evolution, not just for a concentrating subsequence.

For a nondegenerate quartic ground component, the inverse phase operator defines dual translation coordinates in which conserved momentum determines the leading drift. At zero momentum the drift becomes quadratic in the perturbation, improving the exponent. At nonzero momentum the signed main term in moving virial is favorable, so a weaker improvement survives without imposing $P = 0$.

Reading guide. Section 1 gives explicit conserved-spectrum bounds. Section 2 establishes the Fourier and interaction estimates and the concrete local flow. Section 3 proves the general mean bound, and Section 4 proves the conditional momentum refinement. The appendices supply scalar compactness, phase coercivity, quantitative polarization, and an exact obstruction to a stronger instantaneous current bound.

1 Conserved occupations and explicit critical bounds

The local critical nonlinearity admits an explicit connection between scalar sharp interpolation and the entire conserved occupation spectrum. The auxiliary Gaussian variables below occur only in the proof; the initial data and the evolution remain deterministic.

Let $\mathcal{H}$ be a separable complex Hilbert space and suppose the scalar inequality is

$$\int |f|^{2(1+r)} \le (1+r) M_c^{-r} M(f)^r A(f), \qquad r = m/N > 0. \tag{1}$$

The positive scalar Fourier multiplier defining A may be anisotropic. For nonzero $U \in H^{m/2}(\mathbb{R}^N; \mathcal{H})$ set

$$\mathcal{F}(U) = \int \|U\|_{\mathcal{H}}^{2(1+r)}, \qquad \Gamma = \int |U(x)\rangle\langle U(x)|\, dx, \qquad \Gamma_A = \int a(\xi) |\widehat{U}(\xi)\rangle\langle \widehat{U}(\xi)|\, d\xi.$$

Write $M = \operatorname{Tr}\Gamma$, $A = \operatorname{Tr}\Gamma_A$, $\mu = \|\Gamma\|_{\mathrm{op}}$, and list the positive eigenvalues of Γ as $\lambda_1 = \mu \ge \lambda_2 \ge \cdots$. Euler's gamma function is denoted by G in this section, to distinguish it from the correlation operator.

Theorem 1.1 (Explicit bound from the occupation spectrum)**.** *Let ξ_j be independent exponential random variables of mean one. Define*

$$\mathcal{B}_r(\Gamma) = \frac{1}{\mathrm{G}(r+2)} \mathbb{E}\left[\xi_1 \left(\sum_j \lambda_j \xi_j \right)^r \right]. \tag{2}$$

Then

$$\mathcal{F}(U) \le (1+r) M_c^{-r} \mathcal{B}_r(\Gamma) A(U). \tag{3}$$

The coefficient depends only on the spectrum, is homogeneous of degree r, and satisfies $\mathcal{B}_r(\Gamma) \le M^r$. Equality in this last inequality holds precisely when Γ has rank one.

Proof. First suppose the internal rank is finite and diagonalize Γ. For independent standard circular complex Gaussian variables z_j, put $f_z = \sum_j z_j u_j$. Pointwise, f_z has Gaussian variance $\rho_U = \|U\|^2$, so

$$\mathbb{E}\mathcal{F}(f_z) = \mathrm{G}(r+2)\mathcal{F}(U).$$

In the chosen basis $M(f_z) = S := \sum_j \lambda_j |z_j|^2$. Apply (1) and average. Averaging over the phases of the z_j removes off-diagonal entries of the kinetic Gram matrix, giving

$$\mathrm{G}(r+2)\mathcal{F}(U) \le (1+r) M_c^{-r} \sum_j A(u_j) b_j, \qquad b_j = \mathbb{E}[\xi_j S^r]. \tag{4}$$

If $\lambda_i \geq \lambda_j$, exchange ξ_i and ξ_j in one copy of this expectation. With S' denoting the exchanged sum,

$$b_i - b_j = \tfrac{1}{2}\mathbb{E}[(\xi_i - \xi_j)(S^r - (S')^r)] \geq 0,$$

because $S - S' = (\lambda_i - \lambda_j)(\xi_i - \xi_j)$. Thus b_1 is maximal, proving (3). Hölder's inequality and convex Jensen at exponent $r+1$ give

$$\mathbb{E}[\xi_1 S^r] \leq (\mathbb{E}\xi_1^{r+1})^{1/(r+1)}(\mathbb{E}S^{r+1})^{r/(r+1)} \leq \mathrm{G}(r+2)M^r.$$

If a second eigenvalue is positive, S is not proportional to ξ_1 almost surely, and Hölder is strict. In rank one the moment is exactly $M^r\mathrm{G}(r+2)$.

For infinite rank, truncate in the eigenbasis of Γ. The truncated fields converge in $H^{m/2}$ and in the interaction space. Their moments increase to (2) by monotone convergence and remain bounded by M^r. This proves the full result. In particular no identity-covariance Gaussian vector in infinite-dimensional $\mathcal{H}$ is assumed to exist. □

1.1 Explicit quantitative polarization

For $X = \Gamma/M$ and $Y = \Gamma_A/A$, both positive of trace one, the normalized distance to a fixed internal direction is exactly

$$\mathfrak{p}(U) := \inf_{\|\zeta\|=1,q} \left\{ \frac{M(U-\zeta q)}{M} + \frac{A(U-\zeta q)}{A} \right\} = 2 - \lambda_{\max}(X+Y). \tag{5}$$

Indeed the pointwise scalar projection minimizes the combined quadratic norm for each ζ.

Theorem 1.2 (Mass and kinetic alignment, $r \geq 1$)**.** *For complex Hilbert-valued fields,*

$$\frac{\mathcal{F}(U)}{(1+r)(M/M_c)^r A} \leq \frac{1 + r\,\mathrm{Tr}(XY)}{r+1}, \tag{6}$$

$$\mathfrak{p}(U) \leq \frac{2(r+1)}{r}\,\mathfrak{d}(U), \qquad \mathfrak{d}(U) = 1 - \frac{\mathcal{F}(U)}{(1+r)(M/M_c)^r A}. \tag{7}$$

Consequently

$$\mathcal{F}(U) \leq (1+r)(M/M_c)^r \left[1 - \frac{r}{r+1}\left(1 - \frac{\mu}{M}\right)\right] A. \tag{8}$$

For real Hilbert-valued fields with real-even dispersion, the respective factors in (6) *and* (7) *are*

$$\frac{1 + 2r\,\mathrm{Tr}(XY)}{2r+1}, \qquad \frac{2r+1}{r}.$$

Proof. For $r \geq 1$, Jensen gives $S^r \leq M^{r-1}\sum_i \lambda_i \xi_i^r$. Hence

$$\frac{b_j}{\mathrm{G}(r+2)} \leq M^{r-1}\left[\lambda_j + \frac{M - \lambda_j}{r+1}\right].$$

Substitution into (4) proves (6). Since

$$\lambda_{\max}(X+Y) \geq \frac{\mathrm{Tr}((X+Y)^2)}{\mathrm{Tr}(X+Y)} \geq 2\,\mathrm{Tr}(XY),$$

where the second step is $\mathrm{Tr}((X-Y)^2) \geq 0$, (5) proves (7). Also $\mathrm{Tr}(XY) \leq \|X\|_{\mathrm{op}} = \mu/M$, proving (8). For real Gaussian projections use $\xi_j = g_j^2$ with standard real g_j. Their moments satisfy

$$a_q = \mathbb{E}\xi_j^q = \frac{2^q \mathrm{G}(q+1/2)}{\sqrt{\pi}}, \qquad a_{r+1} = (2r+1)a_r.$$

The same calculation gives the stated real constants. Finite-rank approximation proves all assertions in infinite dimension. □

A scalar complex field is a real two-component field $(\operatorname{Re} u, \operatorname{Im} u)$. Therefore, for $r \geq 1$, there is a constant phase θ and a real scalar q such that

$$\frac{M(u - e^{i\theta}q)}{M(u)} + \frac{A(u - e^{i\theta}q)}{A(u)} \leq \frac{2r+1}{r}\mathfrak{d}(u). \tag{9}$$

This supplies the phase-smallness input for near-critical quartic motion in $N \leq 4$ directly from sharp scalar interpolation. The real covariance used for this instantaneous estimate is generally not conserved. The complex internal covariance in the following dynamical result is conserved.

1.2 Consequences for the actual evolution

Consider the same common scalar system

$$i\partial_t U = (a+b)(D)U - \|U\|^{2r}U, \qquad b \geq 0, \qquad E = \frac{A+B}{2} - \frac{\mathcal{F}}{2(1+r)}.$$

Assume its conservative energy-space local flow and continuation property. All pairwise component inner products, and therefore Γ, are conserved. To see this, use the common scalar self-adjoint Hamiltonian $H_U = (a+b)(D) - \|U\|^{2r}$. For smooth approximations the equations give

$$\frac{d}{dt}\int u_j\overline{u_k} = -i\int (H_U u_j)\overline{u_k} + i\int u_j\overline{H_U u_k} = 0.$$

The quadratic-form or strong-approximation interpretation passes this identity to the energy-space flow. In infinite internal dimension, all matrix elements are conserved and determine the positive trace-class operator Γ; continuity also follows from $\|\Gamma(U) - \Gamma(V)\|_1 \leq (\|U\|_2 + \|V\|_2)\|U - V\|_2$.

Theorem 1.3 (Global bounded flow from conserved occupations)**.** *If $\mathcal{B}_r(\Gamma(U_0)) < M_c^r$, the solution is global in both time directions, and*

$$\sup_t A(U(t)) \leq \frac{2E(U_0)}{1 - \mathcal{B}_r(\Gamma(U_0))/M_c^r}, \qquad \sup_t B(U(t)) \leq 2E(U_0).$$

In particular any fixed normalized occupation spectrum of rank greater than one has an explicitly specified global mass interval extending strictly above M_c. At mass M_c, every orbit with unbounded energy norm lies in one fixed internal line for its entire maximal evolution.

Proof. Theorem 1.1 and conservation of Γ yield

$$E(U(t)) \geq \frac{1}{2}\left[1 - \frac{\mathcal{B}_r(\Gamma(U_0))}{M_c^r}\right] A(U(t)) + \frac{1}{2}B(U(t)).$$

Conservation and continuation prove the bounds and global existence. Homogeneity and strictness of $\mathcal{B}_r(\Gamma) < M^r$ give the above-critical interval. At $M = M_c$, unboundedness forces rank one. The fixed range of the conserved Γ then contains $U(t,x)$ almost everywhere at every time. □

Proposition 2.6 verifies these flow assumptions for $m = 2k \geq 4$ and $N = 1, 2, 3$, also for countably many components. In particular it applies to quartic dispersion in these dimensions. No randomness of the initial field, radial symmetry, or finite spatial moment is required.

The spectral moment is directly computable. For integer r, with h_r the complete homogeneous symmetric polynomial,

$$\mathcal{B}_r(\Gamma) = \frac{h_r(\lambda_1, \lambda_1, \lambda_2, \ldots)}{r+1}.$$

In particular,

$$\mathcal{B}_2(\Gamma) = \frac{M^2 + \operatorname{Tr}\Gamma^2 + 2M\mu + 2\mu^2}{6}. \tag{10}$$

For k equal occupations $\lambda_j = M/k$,

$$\frac{\mathcal{B}_r(\Gamma)}{M^r} = \frac{\mathrm{G}(k+r+1)}{k^r \mathrm{G}(k+1)\mathrm{G}(r+2)}.$$

Thus in the two-dimensional quartic system, two equal orthogonal occupations give $\mathcal{B}_2 = M^2/2$: every such H^2 datum with $M < \sqrt{2}M_c$ has globally bounded evolution. Formula (10) retains information beyond M and μ. For $r \geq 1$, at fixed M and λ_1, averaging any two of the remaining occupations cannot increase $\mathcal{B}_r$, by convexity and permutation symmetry of its defining expectation.

These estimates retain the sharp scalar endpoint, but no assertion of optimality at each higher-rank spectrum is made. Evaluating that optimal coefficient is a sharp orthonormal kinetic problem. The Gaussian argument above applies to local powers; Hartree interactions produce an additional exchange term under Gaussian averaging, so no Hartree extension is asserted here.

2 Exterior energy for homogeneous dispersion and attraction

Fix $m > 2$, $s = m/2$, and a real even positive elliptic symbol a, m-homogeneous and C^{N+2} away from zero:

$$c_a|\xi|^m \leq a(\xi) \leq C_a|\xi|^m.$$

For $u \in H^s(\mathbb{R}^N)$ write

$$M = \|u\|_2^2, \qquad A = \|a(D)^{1/2}u\|_2^2, \qquad P = \operatorname{Im}\int \overline{u}\nabla u.$$

We use either the local equation

$$iu_t = a(D)u - |u|^{2m/N}u, \qquad p = 2 + 2m/N, \qquad E = \frac{A}{2} - \frac{\|u\|_p^p}{p}, \tag{11}$$

or, when $2 < m < N$, the Hartree equation

$$iu_t = a(D)u - (K * |u|^2)u, \qquad E = \frac{A}{2} - \frac{H_K(u)}{4}, \quad H_K(u) = \iint K(x-y)|u(x)|^2|u(y)|^2\,dx\,dy. \tag{12}$$

Here K is real, even, C^1 away from zero, homogeneous of degree $-m$, positive definite as a density quadratic form, and

$$|K(x)| + |x||\nabla K(x)| \leq C_K|x|^{-m}.$$

The corresponding sharp inequalities and mass excesses are

$$\begin{aligned} \|u\|_p^p &\leq \frac{p}{2}(M/M_c)^{m/N}A, & d &= (M/M_c)^{m/N} - 1 & &\text{(local)},\\ H_K(u) &\leq 2(M/M_c)A, & d &= M/M_c - 1 & &\text{(Hartree)}. \end{aligned}$$

The dynamical theorem below uses strongly approximable conservative H^s solutions. No local theory for arbitrary symbols and dimensions is silently included in that solution convention. Positive definiteness here is strict on nonzero real densities in the interaction space, so its square root is a strictly convex Hilbert norm. Mere nonnegative semidefiniteness is not the assumed hypothesis.

The exterior kinetic density in both equations is $|a(D)^{1/2}u|^2$. Convexity of a is unnecessary: all anisotropic contributions can be confined to an annulus, with an error tending to zero. We prove that fact directly by Fourier estimates.

2.1 An exterior kinetic term without convexity

Put $b = \sqrt{a}$ and, for $\xi \neq 0$, define the bounded degree-zero symbols

$$B_{jk}(\xi) = \frac{\partial_j a(\xi)\xi_k}{a(\xi)}.$$

Euler's identity gives $\sum_j B_{jj} = m$. These are real even symbols; no sign is assigned to individual entries.

Lemma 2.1 (Cutoff commutations)**.** *For a fixed smooth compact cutoff θ, $\theta_R(x) = \theta(x/R)$, and $R \geq 1$,*

$$\|[B_{jk}(D), \theta_R]b(D)u\|_2 \leq CR^{-1}\|u\|_{H^s}, \tag{13}$$

$$\|[b(D), \theta_R]u\|_2 \leq CR^{-1}\|u\|_{H^s}. \tag{14}$$

The same estimates hold for constant additions to θ_R.

Proof. Write $h = \xi - \zeta$. The angular derivative bound for B_{jk}, its boundedness, and $s > 1$ give

$$|B_{jk}(\xi) - B_{jk}(\zeta)|b(\zeta) \leq C|h||\zeta|^{s-1}.$$

For $|h| \leq |\zeta|/2$ use its homogeneous derivative of degree -1; otherwise use $|\zeta| \leq 2|h|$. Young's convolution inequality and $\||h|\widehat{\theta}_R(h)\|_1 = O(R^{-1})$ prove (13). Similarly,

$$|b(\xi) - b(\zeta)| \leq C|h|(|\zeta| + |h|)^{s-1}$$

gives an upper bound $CR^{-1}\|u\|_{H^{s-1}} + CR^{-s}\|u\|_2$ for (14). Constants commute. □

Choose a radial $h_0 \in C_c^\infty$ with $0 \leq h_0 \leq 1$, equal to one on B_1 and zero outside B_2. Let

$$f_R(x) = xh_0(x/R), \qquad L_R = f_R \cdot \nabla + \tfrac{1}{2}\operatorname{div} f_R, \qquad C_R = [a(D), L_R].$$

Lemma 2.2 (Factorization of the principal form)**.** *For $g = b(D)u$ and $R \geq 1$,*

$$\langle u, C_R u\rangle = \operatorname{Re}\sum_{j,k}\langle g, (\partial_j f_{R,k})B_{jk}(D)g\rangle + O(R^{-1}\|u\|_{H^s}^2). \tag{15}$$

In particular,

$$\langle u, C_R u\rangle \leq mA(u) - m\int_{|x|>2R}|g|^2 + C\int_{R/2<|x|<4R}|g|^2 + CR^{-1}\|u\|_{H^s}^2. \tag{16}$$

All constants are unchanged by translation of the cutoffs.

Proof. The Fourier kernel of C_R, with $h = \xi - \zeta$, is

$$\frac{i}{2}\,[a(\xi) - a(\zeta)](\xi + \zeta)_k \widehat{f}_{R,k}(h).$$

The kernel of $\sum_{j,k}(\partial_j f_{R,k})q_{jk}(D)$, where $q_{jk}(\zeta) = \partial_j a(\zeta)\zeta_k$, is

$$ih_j \widehat{f}_{R,k}(h)\partial_j a(\zeta)\zeta_k.$$

Their difference is bounded by

$$C|h|^2(|\zeta| + |h|)^{m-1}|\widehat{f}_R(h)|. \tag{17}$$

This is second-order Taylor expansion: the second derivatives of a extend continuously at zero because $m > 2$. To bound the form on H^s, divide the kernel by $\langle\xi\rangle^s\langle\zeta\rangle^s$. The elementary bound

$$\frac{(|\zeta| + |h|)^{m-1}}{\langle\xi\rangle^s\langle\zeta\rangle^s} \le C\langle h\rangle^{m-1}$$

follows by splitting $|\zeta| \le 2|h|$ and its complement. Since $\widehat{f_R}(h) = R^{N+1}\widehat{f_1}(Rh)$, Schur's test bounds (17) by CR^{-1} as a form $H^s \times H^s \to \mathbb{C}$.

Replacing $q_{jk}(D)$ by the $b(D)$ sandwich changes its kernel by

$$[b(\zeta) - b(\xi)]b(\zeta)B_{jk}(\zeta)\widehat{\partial_j f_{R,k}}(h).$$

Its magnitude is at most $C|h|(|\zeta| + |h|)^{m-1}|\widehat{\partial_j f_{R,k}}(h)|$. The same weighted Schur estimate, now using the scale-invariant coefficient $\partial_j f_{R,k}$, is $O(R^{-1})$. Taking real parts proves (15).

Write

$$\partial_j f_{R,k} = h_0(x/R)\delta_{jk} + c_{jk,R}.$$

The coefficients $c_{jk,R}$ are uniformly bounded and supported on $R < |x| < 2R$. The identity $\sum_j B_{jj} = m$ makes the diagonal part exactly $m\int h_0(x/R)|g|^2$. Choose θ_R supported on $R/2 < |x| < 4R$ and equal to one on the support of every $c_{jk,R}$. For $c = c_{jk,R}$ and $B = B_{jk}(D)$,

$$\langle g, cBg\rangle = \langle\theta_R g, cB(\theta_R g)\rangle + \langle\theta_R g, c[B, 1-\theta_R]g\rangle.$$

The first term is bounded by $C\|\theta_R g\|_2^2$, and Lemma 2.1 bounds the second by $CR^{-1}\|u\|_{H^s}^2$. Finally $h_0 \le 1$ and $h_0 = 0$ outside $2R$ give (16). No pointwise principal-symbol sign has been used. □

2.2 Localized mass and moving virial identities

Lemma 2.3 (Localized mass transport)**.** *For a real smooth compact cutoff $\chi_R(x) = \chi((x-z)/R)$, $R \ge 1$, and $v, w \in H^s$,*

$$|\langle v, [a(D), \chi_R]w\rangle| \le CR^{-1}\|v\|_{H^s}\|w\|_{H^s}. \tag{18}$$

If at a given time $u = e^{i\theta}(X + iY)$, with X, Y real, $M(Y) \le CdM$, $A(Y) \le CdA(u)$, and the mass is in a fixed compact positive interval, then for $RA(u)^{1/m} \ge 1$,

$$\left|\frac{d}{dt}\int \chi((x-z)/R)|u|^2\right| \le C\sqrt{d}\,R^{-1}A(u)^{(m-1)/m}. \tag{19}$$

The center z in this statement is fixed during differentiation.

Proof. The Fourier kernel of the commutator is $[a(\xi) - a(\zeta)]\widehat{\chi}_R(\xi - \zeta)$. The mean-value bound

$$|a(\xi) - a(\zeta)| \le C|\xi - \zeta|(|\zeta| + |\xi - \zeta|)^{m-1}$$

and the weighted Schur estimate in Lemma 2.2 give (18). The nonlinear potential in both equations is real and cancels from the mass derivative. Since a is real even and χ_R real, the real-real and imaginary-imaginary terms make no contribution; only a bilinear expression in X, Y remains. Normalize $A(u) = 1$ by critical scaling, apply the commutator bound and $\|Y\|_{H^s} \le C\sqrt{d}$, then restore the scale. The temporal scale contributes $A(u)$, while the cutoff radius becomes $RA(u)^{1/m}$, giving (19). The chosen constant phase is never differentiated. □

For a locally absolutely continuous center $z(t)$, define

$$J_R(t) = 2\,\mathrm{Im}\int f_R(x - z(t)) \cdot \overline{u}\nabla u.$$

For smooth solutions, direct differentiation gives

$$J_R' = 2\langle u, C_R^{z(t)} u\rangle + \mathcal{N}_R - 2z'(t)\cdot \int Df_R(x-z(t))\,\mathrm{Im}(\overline{u}\nabla u). \tag{20}$$

The vector field $f_R = xh_0(x/R)$ has symmetric derivative, so this matrix convention is unambiguous. The nonlinear term is

$$\mathcal{N}_R = -\frac{2(p-2)}{p}\int \mathrm{div}\, f_R(x-z)|u|^p \quad \text{for the local power,}$$

and

$$\mathcal{N}_R = \iint |u(x)|^2|u(y)|^2[f_R(x-z)-f_R(y-z)]\cdot\nabla K(x-y)\,dx\,dy \quad \text{for Hartree.}$$

Their full-dilation values combine with $2mA$ to give $4mE$. The motion term equals $-2z'\cdot P$ plus an error bounded by

$$C|z'|\int_{|x-z|>R}|u||\nabla u|, \tag{21}$$

because $Df_R = I$ inside B_R. Also, for mass in a fixed compact interval,

$$|J_R(t)| \le CRA(u(t))^{1/m}. \tag{22}$$

Indeed $\|f_R\|_\infty \le CR$ and interpolation bounds $\|u\|_2\|\nabla u\|_2$ by $CM^{1-1/m}A^{1/m}$. All identities are understood in integrated form for the stated strongly approximable solution class; the Fourier form estimates and the interaction estimates below justify passage to the limit.

2.3 Exterior interaction and current estimates

The preceding estimates were stated on a bounded H^s ball. Their homogeneity gives precise instantaneous kinetic dependence. Fix the mass in a compact positive interval, put $\lambda = A(u)^{1/m}$, and assume $R\lambda \ge 1$. Rescaling to $A = 1$ in Lemma 2.2 replaces its remainder by

$$CR^{-1}A(u)^{(m-1)/m}. \tag{23}$$

For $\zeta_R = 0$ on $B_{R/4}$ and $\zeta_R = 1$ outside $B_{R/2}$, Lemma 2.1 similarly gives

$$\|[a(D)^{1/2},\zeta_R]u\|_2^2 \le CR^{-2}A(u)^{(m-2)/m}. \tag{24}$$

These follow by unitary critical spatial rescaling; the constants do not require an upper bound on $A(u)$.

Fix a center z and write $g = a(D)^{1/2}u$,

$$K_R = \int_{|x-z|>8R}|g|^2, \qquad H_R = \int_{R/4<|x-z|<8R}|g|^2, \qquad \varepsilon_R = R^{-2}A(u)^{(m-2)/m}.$$

If $\int_{|x-z|>R/4}|u|^2 \le \mu$, then

$$M(\zeta_R u) \le \mu, \qquad A(\zeta_R u) \le C(K_R + H_R + \varepsilon_R). \tag{25}$$

Sharp interpolation and interpolation between orders zero and s give

$$\int_{|x-z|>R}|u|^p \le C\mu^{m/N}(K_R + H_R + \varepsilon_R), \tag{26}$$

$$\int_{|x-z|>R}|u||\nabla u| \le C\mu^{1-1/m}(K_R + H_R + \varepsilon_R)^{1/m}. \tag{27}$$

For the current bound, apply $\|\nabla w\|_2 \le C\|w\|_2^{1-2/m}A(w)^{1/m}$ to $w = \zeta_R u$, which equals u on the indicated exterior region.

Lemma 2.4 (Hartree virial tail)**.** *In the Hartree setting the difference between the nonlinear localized virial expression and its full dilation value $-mH_K(u)$ obeys*

$$|\textit{Hartree defect}| \le CR^{-m}M^2 + C\mu(K_R + H_R + \varepsilon_R). \tag{28}$$

Proof. For $\rho = |u|^2$ the symmetrized nonlinear expression is

$$\iint \rho(x)\rho(y)[f_R(x-z) - f_R(y-z)] \cdot \nabla K(x-y)\,dx\,dy.$$

The full dilation vector gives $-mH_K(u)$ by homogeneity. The difference vanishes if both points lie in $B(z,R)$, and the uniform Lipschitz bound for f_R bounds its absolute integrand by $C\rho(x)\rho(y)|x-y|^{-m}$. By symmetry take $|x-z| > R$. If $|y-z| < R/2$, then $|x-y| \ge R/2$, and this contribution is at most $CR^{-m}M^2$. Otherwise both points are outside $B(z,R/2)$. For $w = \zeta_R u$, HLS and elliptic interpolation give

$$\iint |x-y|^{-m}|w(x)|^2|w(y)|^2\,dx\,dy \le C\|w\|^4_{4N/(2N-m)} \le CM(w)A(w).$$

Use (25). Thus the near-diagonal exterior interaction sees only the small mass μ, while the separated core–tail interaction vanishes with R. □

Lemma 2.5 (Hartree weak energy splitting)**.** *Suppose $u_n \rightharpoonup v$ in H^s and $u_n \to v$ in L^2_{loc}. For $w_n = u_n - v$,*

$$H_K(u_n) = H_K(v) + H_K(w_n) + o(1), \qquad E(u_n) = E(v) + E(w_n) + o(1).$$

Proof. Set $q = 4N/(2N-m)$, strictly between 2 and the H^s Sobolev exponent. Thus $w_n \to 0$ in L^q_{loc} and is bounded globally in L^q. Approximate v by smooth compact functions to obtain $\mathrm{Re}(\overline{v}w_n) \to 0$ in $L^{q/2}$. HLS removes all corresponding terms in the density expansion. It remains to show $\iint K(x-y)|v(x)|^2|w_n(y)|^2\,dx\,dy \to 0$. The potential of a smooth compact approximation to v is bounded because $m < N$, and tends to zero at infinity by the Riesz bound. Local strong L^2 convergence and bounded mass of w_n make this pairing tend to zero. HLS controls the approximation error. The kinetic quadratic form splits by weak convergence, proving the energy identity. □

2.4 A concrete conservative local flow

Proposition 2.6 (Integer energy regularity in dimensions one to three)**.** *Let $m = 2k$ with integer $k \ge 2$, $N = 1, 2, 3$, and let $\mathcal{H}$ be any separable complex Hilbert space. For every symbol a above, let b be a common real measurable symbol satisfying $0 \le b(\xi) \le C_b(1 + |\xi|^{2k})$. The total-density system*

$$i\partial_t U = [a(D) + b(D)]U - \|U\|^{4k/N}_{\mathcal{H}} U$$

has a unique maximal $C_t H^k_x(\mathcal{H})$ solution for every $H^k(\mathcal{H})$ datum. The flow conserves

$$M = \int \|U\|^2_{\mathcal{H}}, \qquad P = \mathrm{Im}\int \langle U, \nabla U\rangle_{\mathcal{H}},$$
$$E_b = \frac{1}{2}\int (a+b)(\xi)\|\widehat{U}(\xi)\|^2_{\mathcal{H}}\,d\xi - \frac{1}{\alpha+2}\int \|U\|^{\alpha+2}_{\mathcal{H}}, \qquad \alpha = 4k/N,$$

and has the usual H^k continuation alternative. The local estimates are independent of the internal dimension. For the homogeneous case $b = 0$, the localized mass and virial identities also pass by strong approximation. In particular the scalar dynamical conclusions below need no additional local-theory assumption throughout this integer-order class, which includes every positive elliptic even homogeneous polynomial of degree $2k$. No homogeneous virial identity is asserted for a general nonzero b.

Proof. Write $\alpha = 4k/N > k$. On the underlying real Hilbert space, the radial map $F(Z) = \|Z\|_{\mathcal{H}}^{\alpha} Z$ is Fréchet C^{k+1}. Its derivatives obey, for $1 \le j \le k+1$,

$$\|D^j F(Z)\|_{\mathrm{op}} \le C_{\alpha,j} \|Z\|_{\mathcal{H}}^{\alpha+1-j},$$

with constants independent of $\mathcal{H}$; the derivatives extend continuously at zero because $\alpha + 1 - j > 0$. These bounds follow by differentiating the Hilbert norm away from zero: each term is a product of inner products, vectors, and a power of the norm. Since $k > N/2$, the Hilbert-valued Sobolev embedding and integer chain/product estimates, applied to

$$F(U) - F(V) = \int_0^1 DF(V + \tau(U - V))(U - V)\, d\tau,$$

show that $F : H^k(\mathcal{H}) \to H^k(\mathcal{H})$ is locally Lipschitz. The estimates use operator norms of the displayed derivatives and pointwise Hilbert norms, so their constants do not grow with the internal dimension. They extend from finite-component smooth fields by orthogonal projection and density. The common scalar group $e^{-it[a(D)+b(D)]}$ is strongly continuous and unitary on $H^k(\mathcal{H})$. Duhamel contraction gives existence, uniqueness, continuous dependence, and the blowup alternative.

For conservation and approximation, use the self-adjoint Fourier projection Π_n onto $|\xi| \le n$ and solve

$$i\partial_t U_n = [a(D) + b(D)]U_n - \Pi_n F(U_n), \qquad U_n(0) = \Pi_n U_0, \qquad U_n = \Pi_n U_n.$$

This projected Hamiltonian flow conserves the restrictions of M, P, E_b: Π_n commutes with translations and the linear dispersion, and its range is invariant under constant phase. The local Lipschitz bounds are uniform in n, so Duhamel and Gronwall give $U_n \to U$ in $C_t H_x^k(\mathcal{H})$ on every compact subinterval of the lifespan. All three conserved functionals are continuous in $H^k(\mathcal{H})$, using the stated growth bound for b, so conservation passes to the limit. When $b = 0$, the same approximation proves the integrated localized identities: the linear commutators are continuous H^k forms by the Fourier bounds above, which also hold componentwise with dimension-independent constants. The nonlinear projection error vanishes because $F(U_n) \to F(U)$ in H^k and the fixed-radius first-order virial operator is bounded from H^k to L^2. No dispersive decay estimate is required. □

3 Persistent amplification for homogeneous dispersion

The exterior commutator estimates yield a stronger statement than one kinetic excursion: a lower long-time average in actual physical time, without a global kinetic bound. Let $m > 2$, $s = m/2$, and let a satisfy the positive elliptic, real-even, homogeneous symbol assumptions above. Write $A(u) = \|a(D)^{1/2}u\|_2^2$, $M(u) = \|u\|_2^2$ and $P(u) = \mathrm{Im} \int \overline{u} \nabla u$.

We treat both the local equation $iu_t = a(D)u - |u|^{2m/N}u$ and the Hartree equation $iu_t = a(D)u - (K * |u|^2)u$ under the preceding Hartree kernel assumptions: $2 < m < N$, a real-even, degree-$-m$ homogeneous C^1 kernel away from zero, positive-definite density form, and $|K(x)| + |x||\nabla K(x)| \le C|x|^{-m}$. Put

$$\kappa = 1 + m/N \quad \text{(local)}, \qquad \kappa = 2 \quad \text{(Hartree)}.$$

Their sharp energy inequalities have the common form

$$E(u) = A(u)/2 - \mathcal{F}(u)/(2\kappa), \qquad \mathcal{F}(u) \le \kappa (M(u)/M_c)^{\kappa-1} A(u).$$

Here $\mathcal{F} = \|u\|_{2\kappa}^{2\kappa}$ locally and is the Hartree interaction otherwise.

Theorem 3.1 (Lower long-time kinetic average)**.** *For each fixed equation above there are $d_0, c > 0$ such that every forward-global conservative strong H^s solution, for which the localized identities pass by strong energy-space approximation, with*

$$P = 0, \qquad E = -e < 0, \qquad 0 < d = (M/M_c)^{\kappa-1} - 1 < d_0$$

satisfies

$$\boxed{\liminf_{T\to\infty}\frac{1}{T}\int_0^T A(u(t))\,dt \geq ced^{-\alpha_m}, \qquad \alpha_m = 1+\frac{m}{2(m-1)}.} \tag{29}$$

No uniform kinetic bound, radial symmetry, finite spatial moment, convexity of a, or real-sector linearized nondegeneracy is assumed. The theorem specifies its solution class; it does not assert local well-posedness for every symbol and dimension.

Proof. Write $A(t) = A(u(t))$, $\delta = M - M_c$, $\gamma = (m-1)/m$. Sharp interpolation gives $A(t) \geq a_* := 2e/d > 0$. Real polarization supplies, separately at each time, a phase with $u = e^{i\theta}(X+iY)$, X, Y real, and $M(Y) \leq CdM$, $A(Y) \leq CdA(t)$. The nonlinear potential is real in both models and cancels from the localized mass derivative. Apply the normalized Fourier mass-commutator bound after scaling A to one. For $RA(t)^{1/m} \geq 1$ it gives

$$\left|\frac{d}{dt}\int \chi((x-z)/R)|u|^2\right| \leq C\sqrt{d}R^{-1}A(t)^{\gamma}. \tag{30}$$

No time derivative of the chosen phase is taken.

Center and uniform mass tail. Fix $\varepsilon_0 = M_c/32$ and take d_0 small. Normalized near-optimizer compactness gives a fixed $\overline{R}$ and centers $c(t)$ with

$$\int_{|x-c(t)|\leq \overline{R}A(t)^{-1/m}} |u(t,x)|^2\,dx \geq M_c - \varepsilon_0.$$

Choose $R_0 \geq 1$ containing these balls and satisfying $R_0 \geq a_*^{-1/m}$. Introduce $s(t) = \int_0^t A(\tau)^{\gamma}d\tau$. Two centers separated by more than $4R_0$ force a cutoff mass change of at least $M_c - 3\varepsilon_0$. Equation (30) excludes this when $|s(t) - s(t')| \leq \tau R_0/\sqrt{d}$, with fixed small $\tau > 0$. Interpolating sampled centers at these intrinsic-time intervals gives a locally absolutely continuous $b(t)$ with

$$|b(t) - c(t)| \leq 8R_0, \qquad |b'(t)| \leq C\sqrt{d}A(t)^{\gamma} \quad \text{a.e.} \tag{31}$$

The speed constant is independent of R_0 and e/d.

For every $\eta > 0$ there is $R_\eta < \infty$ such that

$$\sup_{t\geq 0}\int_{|x-b(t)|>R_\eta} |u(t,x)|^2\,dx \leq \delta + \eta. \tag{32}$$

Here is the compactness argument without a kinetic upper bound. Suppose bad times t_k and radii $R_k \to \infty$ exist. The bad mass lies also outside $B(c(t_k), R_k - 8R_0)$. Set $A_k = A(t_k)$, $\ell_k = A_k^{-1/m}$ and $v_k(y) = \ell_k^{N/2}u(t_k, c(t_k) + \ell_k y)$. Then $A(v_k) = 1$, $M(v_k) = M$, $E(v_k) = -e/A_k \in [-d/2, 0)$, and a fixed ball contains $M_c - \varepsilon_0$ mass. A weak H^s limit v, with strong local L^2 convergence, therefore has $M(v) \geq M_c - \varepsilon_0 > 0$. The normalized bad radius tends to infinity, since $(R_k - 8R_0)\ell_k^{-1} \geq (R_k - 8R_0)a_*^{1/m} \to \infty$. Consequently $M(v) \leq M_c - \eta$. For $w_k = v_k - v$, mass and energy split:

$$E(v_k) = E(v) + E(w_k) + o(1), \qquad M(w_k) \leq \delta + \varepsilon_0 + o(1) < M_c.$$

Locally the nonlinear splitting is Brezis–Lieb. For Hartree, the Riesz bound and HLS imply that $v\overline{w_k}$ vanishes in the density space, while $|w_k|^2$ converges weakly to zero there. First approximate v by bounded compactly supported functions and use local compactness; then use uniform Sobolev and HLS bounds. This removes all mixed interaction terms. Sharp interpolation gives $E(w_k) \geq 0$ eventually and $E(v) > 0$, because $v \neq 0$ is strictly subcritical, contradicting $E(v_k) < 0$. This proves (32).

Variable-scale virial. Let $g = a(D)^{1/2}u$ and use the compact vector field f_R of the exterior commutator lemma. Define

$$J_R = 2\,\mathrm{Im}\int f_R(x-b(t))\cdot\overline{u}\nabla u, \quad K_R = \int_{|x-b|>8R}|g|^2, \quad H_R = \int_{R/4<|x-b|<8R}|g|^2.$$

Scaling the normalized exterior commutator and cutoff-energy estimates back to $A(t)$ gives errors $CR^{-1}A^{\gamma}$ and $CR^{-2}A^{(m-2)/m}$. Write

$$q_R(t) = R^{-1}A(t)^{\gamma} + R^{-2}A(t)^{(m-2)/m} + R^{-m}.$$

The last term is needed only for separated Hartree interactions. For $R \geq 4R_\eta$, $R \geq a_*^{-1/m}$ and $\mu = \delta + \eta$, the preceding localized identities give

$$\begin{aligned} J_R' \leq & -4me - 2mK_R + CH_R + Cq_R \\ & + C\mu^{\kappa-1}(K_R + H_R + q_R) \\ & + C|b'|\mu^{1-1/m}(K_R + H_R + q_R)^{1/m}. \end{aligned}$$

The constant center-motion term is $-2b' \cdot P = 0$. Its remainder is bounded using $\|\zeta_R u\|_2 \|\nabla(\zeta_R u)\|_2$ and Fourier interpolation, since $s > 1$. For Hartree, separated inner–outer interactions are $O(R^{-m}M^2)$; the remaining exterior interaction is bounded by $C\mu A(\zeta_R u)$.

Take d_0, η small enough to absorb the nonlinear K_R term. Young's inequality absorbs the current's K_R term and bounds the other parts by $C(H_R + q_R) + C\mu|b'|^{m/(m-1)}$. Equation (31) implies

$$J_R' \leq -4me + CH_R + C\mu d^{m/[2(m-1)]}A(t) + Cq_R(t). \tag{33}$$

The endpoint estimate is $|J_R(t)| \leq CRA(t)^{1/m}$.

Physical-time averaging. Let L be the left side of (29); only finite L needs proof. Choose $T_n \to \infty$ with averages tending to L. There is $t_n \in [T_n - \sqrt{T_n}, T_n]$ with $A(t_n) \leq C(L+1)\sqrt{T_n}$. Therefore $t_n/T_n \to 1$, the averages through t_n tend to L, and $A(t_n)^{1/m}/t_n \to 0$. For fixed $K < \infty$ and large R_*, the annuli at radii $R_j = 64^j R_*$, $0 \leq j < K$, are disjoint, so $\sum_j H_{R_j}(t) \leq A(t)$. For each n choose one radius with average annular energy at most K^{-1} times the average of A and integrate (33). The chosen radii stay bounded for fixed K, R_*, so endpoints divided by t_n vanish. Jensen's inequality bounds the fractional powers of A in q_R. The time limit yields

$$4me \leq CL/K + C\mu d^{m/[2(m-1)]}L + CR_*^{-1}L^{\gamma} + CR_*^{-2}L^{(m-2)/m} + CR_*^{-m}.$$

Let $K \to \infty$, then $R_* \to \infty$, then $\eta \downarrow 0$. Since $\delta = M_c[(1+d)^{1/(\kappa-1)} - 1] \leq Cd$, this proves (29). □

Corollary 3.2 (Repeated amplification and standing-wave instability)**.** *For any real sharp standing wave Q with $M(Q) = M_c$, take $u_d(0) = (1+d)^{1/[2(\kappa-1)]}Q$. Every global solution in the stated class satisfies*

$$\liminf_{T\to\infty} \frac{1}{T}\int_0^T \frac{A(u_d(t))}{A(u_d(0))}\,dt \geq cd^{-m/[2(m-1)]}.$$

Every prescribed finite amplification factor is thus exceeded at arbitrarily late times for small enough d. Under the usual local continuation alternative, each real sharp standing wave is nonlinearly orbitally unstable modulo phase and translation.

Proof. The exact amplitude identity gives $e = dA(u_d(0))/2$. The initial perturbations converge to Q in H^s, whereas a large kinetic excursion leaves a fixed neighborhood of its phase–translation orbit. Finite-time breakdown also forces such an excursion under the continuation alternative. □

If a particular orbit has $B = \sup_t A(t) < \infty$ and $L_* = ced^{-\alpha_m}$, then the set $\{t : A(t) \geq L_*/2\}$ has lower asymptotic density at least $L_*/(2B)$. Without this extra boundedness, the theorem still permits sparse high spikes. For $m = 4$, $\alpha_m = 5/3$ and the perturbation exponent is $2/3$. The result does not by itself exclude every bounded global negative-energy orbit.

4 Momentum-adapted centers and stronger amplification

The preceding amplification theorem requires no nondegeneracy of the real linearized operator. We now show what that additional hypothesis buys dynamically: conserved momentum removes the leading translation speed in a suitable modulation coordinate. The scale and phase equations must be retained for this cancellation to be useful.

Throughout this section $N \in \{1, 2, 3\}$ and

$$iu_t = \Delta^2 u - |u|^{8/N} u, \qquad p = 2 + 8/N.$$

Let Q be a real sharp ground wave satisfying

$$\Delta^2 Q + Q - |Q|^{p-2} Q = 0, \qquad M(Q) = M_c,$$

and put

$$L_- = \Delta^2 + 1 - |Q|^{p-2}, \quad L_+ = \Delta^2 + 1 - (p-1)|Q|^{p-2}, \quad \Lambda = N/2 + x \cdot \nabla.$$

We assume

$$\ker L_+ = \operatorname{span}\{\partial_j Q : 1 \le j \le N\}. \tag{34}$$

The phase coercivity and simplicity of $\ker L_-$ are provided by the sharp variational problem. The weighted regularity of the profile and dual functions needed below follows from this concrete elliptic equation, as we prove next; it is not an additional assumption.

Lemma 4.1 (Exponential regularity of the dual functions)**.** *For some $a > 0$, $e^{a\langle x\rangle} Q \in H^6$. If $L_- Z = f$ on $Q^\perp$, with $f = \partial_j Q$ or $f = \Lambda Q$, then $e^{a'\langle x\rangle} Z \in H^6$ for some $a' > 0$. In particular $\Lambda Q \in H^4$, and the functions Z_j, W below satisfy $\nabla Z_j, \Lambda Z_j, \nabla W, \Lambda W \in L^2$.*

Proof. The H^2 algebra property and the profile equation give $Q \in H^6$. Hence $V = |Q|^{8/N} \in W^{2,\infty}$ and V tends to zero at infinity. For smooth bounded truncations ϕ_R of $a\langle x\rangle$, put $w_R = e^{\phi_R} Q$. The exact identity

$$\operatorname{Re}\langle \Delta^2 Q, e^{2\phi_R} Q\rangle = \|\Delta w_R + |\nabla \phi_R|^2 w_R\|_2^2 - \|2\nabla\phi_R \cdot \nabla w_R + (\Delta\phi_R) w_R\|_2^2$$

differs from $\|\Delta w_R\|_2^2$ by at most $Ca^2 \|w_R\|_{H^2}^2$. The potential is small outside a fixed ball; its weighted integral inside that ball is uniformly bounded. Absorption therefore gives a uniform weighted H^2 bound. The conjugated equation has smooth bounded lower-order coefficients of order at most three. Since multiplication by V preserves H^2, successive elliptic gains give $e^{a\langle x\rangle} Q \in H^6$.

For $L_- Z = f$, choose $0 < a' < a$ small enough and truncate $a'\langle x\rangle$ with $\phi_R \le a'\langle x\rangle$. The same test produces the source term

$$|\langle e^{\phi_R} f, e^{\phi_R} Z\rangle| \le \tfrac{1}{4} \|e^{\phi_R} Z\|_2^2 + C \|e^{a'\langle x\rangle} f\|_2^2.$$

The rightmost norm is finite, and the interior weighted potential is bounded by the fixed unweighted L^2 norm of Z on the ball. Absorption and passage to the limit again give weighted H^2. Moreover $e^{a'\langle x\rangle} f \in H^2$. The conjugated equation then gives the successive gains $H^2 \to H^3 \to H^4 \to H^5 \to H^6$. Polynomially weighted derivatives of lower order follow by reducing the exponential weight. Differentiating the scaled profile equation now also justifies $L_+ \Lambda Q = -4Q$. □

Theorem 4.2 (Quadratic drift and stronger amplification)**.** *Under these hypotheses there are constants $d_Q, C_Q > 0$ and a neighborhood $\mathcal{U}_Q$ of the normalized phase–translation orbit of Q such that the following holds. Let u be a conservative, strongly approximable H^2 solution with the usual continuation property, and suppose*

$$0 < d = \left(\frac{M}{M_c}\right)^{4/N} - 1 < d_Q, \qquad E = -e < 0, \qquad P = 0.$$

Assume the datum, after amplitude normalization to mass M_c and spatial normalization to kinetic energy $A(Q)$, lies in $\mathcal{U}_Q$. Every forward-global such solution satisfies

$$\liminf_{T\to\infty} \frac{1}{T}\int_0^T A(u(t))\,dt \geq c_Q e d^{-7/3}. \tag{35}$$

In particular, if $B = \sup_{t\geq 0} A(u(t)) < \infty$, then

$$e \leq C_Q d^{7/3} B. \tag{36}$$

In particular, for $u_d(0) = (1+d)^{N/8}Q$,

$$\sup_{0\leq t<T_+} \frac{A(u_d(t))}{A(u_d(0))} \geq c_Q d^{-4/3}. \tag{37}$$

Every factor strictly smaller than the right-hand side is attained at a finite time, and at arbitrarily late times if the solution is global. In that case the lower long-time average of $A(u_d(t))/A(u_d(0))$ is also at least $c_Q d^{-4/3}$. No radial condition is imposed on the evolving solution.

Proof. Define

$$Z_j = L_-^{-1}\partial_j Q, \quad T_{ij} = \langle \partial_i Q, Z_j\rangle, \quad c_j = \langle \Lambda Q, Z_j\rangle.$$

Phase coercivity implies $T > 0$. Translating $Q(x)$ to $Q(x+b)$ changes c to $c - Tb$. Choose $b = T^{-1}c$, so that $c = 0$, and define

$$W = L_-^{-1}\Lambda Q, \qquad D = \langle \Lambda Q, W\rangle > 0.$$

Selfadjointness gives

$$\langle \Lambda Q, Z_j\rangle = \langle \partial_j Q, W\rangle = 0. \tag{38}$$

Strict positivity of D follows from phase coercivity and $\Lambda Q \neq 0$; a nonzero L^2 function cannot be homogeneous of degree $-N/2$.

Coercivity and isolation. Sharp interpolation implies $E(f) \geq 0$ on $M(f) = M_c$. Therefore $\langle L_+h, h\rangle \geq 0$ on $Q^\perp$. A zero vector of this constrained form satisfies $L_+h \in \mathbb{R}Q$. Since $L_+\Lambda Q = -4Q$, assumption (34) identifies the constrained null space as $\operatorname{span}\{\Lambda Q, \partial_1 Q, \ldots, \partial_N Q\}$. The decaying potential is form-compact relative to $\Delta^2 + 1$. Consequently, (38) and $T > 0$ yield

$$\langle L_+h, h\rangle + \langle L_-k, k\rangle \geq c_Q(\|h\|_{H^2}^2 + \|k\|_{H^2}^2) - C_Q\langle h, Q\rangle^2 \tag{39}$$

whenever

$$\langle h, Z_j\rangle = \langle h, W\rangle = \langle k, Q\rangle = 0. \tag{40}$$

The same estimate, applied to a nearby exact optimizer with its exact mass and energy constraints, shows that the normalized orbit of Q is isolated. Indeed its perturbation norm is bounded by a higher-order remainder in that same norm and must vanish.

For a negative-energy trajectory of mass parameter d, the normalized sharp-interpolation deficit is at most d. Sharp-optimizer compactness therefore places the normalized trajectory uniformly close to the set of normalized optimizers as $d \to 0$. Choose $\rho > 0$ so that the quotient ball of radius 2ρ about Q's orbit contains no other normalized optimizer orbit. A state at distance ρ from that orbit is at distance at least ρ from every optimizer orbit, by the triangle inequality. This contradicts uniform distance less than $\rho/4$ from the optimizer set for small d. Thus a continuous normalized trajectory starting at distance less than $\rho/2$ cannot leave the ρ tube. Choosing ρ small enough proves that one fixed Q can be used throughout the maximal existence interval.

Exact modulation equations. Write

$$u(t,x) = \lambda(t)^{N/2}e^{i\theta(t)}w(s,y), \qquad w = Q + h + ik, \qquad y = \lambda(t)(x - a(t)), \qquad \frac{ds}{dt} = \lambda^4,$$

with (40). The parameter Jacobian at Q has invertible blocks T, D, M_c. Lemma 4.1 justifies local absolute continuity of the parameters and differentiation of the gauges. Terms involving Λh are paired by integration by parts against ΛZ_j or ΛW. Set

$$b = \lambda_s/\lambda, \qquad v = \lambda a_s, \qquad \alpha = \theta_s - 1,$$

and, for $F(z) = |z|^{p-2}z$, set

$$\mathcal{R} = F(Q + h + ik) - F(Q) - (p-1)|Q|^{p-2}h - i|Q|^{p-2}k.$$

The exact real equations are

$$h_s = L_- k - \operatorname{Im}\mathcal{R} + \alpha k - b\Lambda(Q+h) + v\cdot\nabla(Q+h), \tag{41}$$

$$k_s = -L_+ h + \operatorname{Re}\mathcal{R} - \alpha(Q+h) - b\Lambda k + v\cdot\nabla k. \tag{42}$$

If $\epsilon^2 = \|h\|_{H^2}^2 + \|k\|_{H^2}^2$, then $\|\mathcal{R}\|_2 \le C_Q\epsilon^2$, since $N \le 3$, $p > 4$, and $H^2 \hookrightarrow L^\infty$.

Zero momentum is exactly

$$\langle k, \partial_j Q\rangle = -\langle k, \partial_j h\rangle. \tag{43}$$

Let $(T_h)_{j\ell} = \langle \partial_\ell h, Z_j\rangle$. Differentiate the translation, scale and phase gauges in that order:

$$[(T+T_h)v]_j = \langle k, \partial_j h\rangle + \langle \operatorname{Im}\mathcal{R}, Z_j\rangle - \alpha\langle k, Z_j\rangle + b\langle \Lambda h, Z_j\rangle, \tag{44}$$

$$b[D + \langle \Lambda h, W\rangle] - v\cdot\langle\nabla h, W\rangle = \langle k, \Lambda Q\rangle - \langle \operatorname{Im}\mathcal{R}, W\rangle + \alpha\langle k, W\rangle, \tag{45}$$

$$\alpha[M_c + \langle h, Q\rangle] + b\langle \Lambda k, Q\rangle - v\cdot\langle\nabla k, Q\rangle = -\langle h, L_+ Q\rangle + \langle \operatorname{Re}\mathcal{R}, Q\rangle. \tag{46}$$

This linear system for v, b, α is a small perturbation of one with diagonal blocks T, D, M_c, so $|v| + |b| + |\alpha| \le C_Q\epsilon$. Substitution in (44) improves the translation estimate to

$$|v| \le C_Q\epsilon^2. \tag{47}$$

The phase and scale terms have thus been controlled, not omitted.

Energy and center speed. Put $\delta M = M - M_c = O(d)$. The mass identity gives $2\langle h, Q\rangle = \delta M - \|h\|_2^2 - \|k\|_2^2$. For $S = E + M/2$, Taylor expansion and (39) imply

$$S(w) - S(Q) \ge c_Q\epsilon^2 - C_Q(\delta M)^2 - o(\epsilon^2).$$

But $E(w) = \lambda^{-4}E(u) < 0$, so $S(w) - S(Q) \le \delta M/2$. Shrinking the tube yields

$$\epsilon^2 \le C_Q d, \qquad |a_t| = \lambda^3|v| \le C_Q d\lambda^3. \tag{48}$$

Since $A(w)$ stays comparable to $A(Q)$, (48) gives the scale-sensitive speed

$$|a_t| \le C_Q d A(t)^{3/4}$$

without any upper bound on $A(t)$. A ball of radius $C_Q A(t)^{-1/4}$ about $a(t)$ contains at least $M_c - \varepsilon_0$ mass, where $\varepsilon_0 > 0$ is fixed and small. The bound $A(t) \ge 2e/d$ makes these physical radii uniformly bounded.

For every $\eta > 0$ the mass outside a sufficiently large fixed radius about $a(t)$ is at most $M - M_c + \eta$, uniformly in time. To justify this without a kinetic bound, suppose bad times and radii tend to infinity and normalize each state to kinetic energy one about its center. The bad normalized radii tend to infinity because $A(t) \ge 2e/d$. Weak H^2 compactness gives a nonzero core of mass between $M_c - \varepsilon_0$ and $M_c - \eta$. The remainder is also subcritical. Their split energies have strictly positive sum, in contradiction with the normalized total energies $-e/A(t) \le 0$.

Put $\mu = M - M_c + \eta$. The negative exterior kinetic contribution of Lemma 2.2, with $a(\xi) = |\xi|^4$, absorbs the exterior current by Young's inequality. With H_R denoting the transition-annulus kinetic energy, the resulting estimate is

$$J_R' \le -16e + CH_R + C_Q\mu d^{4/3}A(t) + C[R^{-1}A(t)^{3/4} + R^{-2}A(t)^{1/2} + R^{-4}].$$

Let $L = \liminf_{T\to\infty} T^{-1}\int_0^T A$. If $L = \infty$ the claim follows. Otherwise choose times $t_n \to \infty$ with the averages tending to L and $A(t_n)^{1/4}/t_n \to 0$; the endpoint selection in the preceding actual-time argument applies unchanged. Since $|J_R(t)| \le CRA(t)^{1/4}$, endpoint terms disappear after division by t_n. Among K disjoint geometrically spaced annuli, one has average kinetic energy at most the total average divided by K. Integrate at that radius, then let time tend to infinity, K tend to infinity, the base radius tend to infinity, and η tend to zero. This gives

$$e \le C_Q(M - M_c)d^{4/3}L \le C_Q d^{7/3}L.$$

It proves (35), and (36) follows immediately. For the specified initial data $E = -dA(u_d(0))/2$ and $P = 0$. The average bound gives arbitrarily late crossings for global solutions. Finite-time breakdown makes the kinetic supremum infinite by continuation. Continuity gives the stated finite-time crossings. □

Theorem 4.3 (Arbitrary momentum in the same ground component)**.** *Retain the hypotheses of Theorem 4.2, except that P may now be arbitrary. Every forward-global such solution satisfies*

$$\liminf_{T\to\infty} \frac{1}{T}\int_0^T A(u(t))\,dt \ge c_Q|E|d^{-2}. \tag{49}$$

At $P = 0$, (35) gives the stronger exponent 7/3.

Proof. Component trapping, constrained coercivity and $\epsilon^2 \le C_Q d$ did not use $P = 0$. For general momentum the exact identity replacing (43) is

$$\langle k, \partial_j Q\rangle = -\frac{P_j}{2\lambda} - \langle k, \partial_j h\rangle.$$

The right-hand side of (44) therefore gains $P_j/(2\lambda)$. Since $|P|/\lambda \le C_Q\epsilon$, solving the same coupled equations first to linear order and substituting back gives

$$v = \tfrac{1}{2}T^{-1}(P/\lambda) + O_Q(d), \qquad a_t = \tfrac{\lambda^2}{2}T^{-1}P + O_Q(d\lambda^3). \tag{50}$$

Retain the exact $-2a_t \cdot P$ term in localized virial. Positivity of T gives

$$-2a_t \cdot P \le -c_Q\lambda^2|P|^2 + C_Q d^2\lambda^4.$$

After absorbing exterior kinetic energy, the exterior current costs $C\mu|a_t|^{4/3}$, where $\mu = M - M_c + \eta$. For any fixed small $\varepsilon_1 > 0$, (50) and Young's inequality imply

$$\begin{aligned}\mu|a_t|^{4/3} &\le C_Q\mu\lambda^{8/3}|P|^{4/3} + C_Q\mu d^{4/3}\lambda^4\\ &\le \varepsilon_1\lambda^2|P|^2 + C_{Q,\varepsilon_1}(\mu^3 + \mu d^{4/3})\lambda^4.\end{aligned}$$

The first term is absorbed into the negative momentum contribution. Since λ^4 is comparable to $A(t)$, this leaves

$$J_R' \le -16e + CH_R + C_Q[d^2 + \mu^3 + \mu d^{4/3}]A(t) + C[R^{-1}A(t)^{3/4} + R^{-2}A(t)^{1/2} + R^{-4}].$$

Uniform surplus-tail mass and actual-time endpoint selection remain valid. Averaging disjoint annuli and sending $\eta \to 0$ gives

$$e \le C_Q[d^2 + (M - M_c)^3 + (M - M_c)d^{4/3}]\liminf_{T\to\infty}\frac{1}{T}\int_0^T A(t)\,dt.$$

Because $M - M_c = O(d)$, the bracket is $O(d^2)$. This proves (49). No Galilean invariance is used. □

The conditional exponent 4/3 improves the unconditional exponent 2/3. It still permits a bounded trajectory at a sufficiently large kinetic scale. Reapplying the estimate later leaves d and E unchanged and does not force iteration to infinity. A further dynamical argument must control scale evolution or the quadratic translation terms in (44) over long times.

5 Relation to existing work and the remaining evolution problem

For the pure quartic equation, Boulenger–Lenzmann [1] prove radial negative-energy finite- or infinite-time blowup, and Bonheure–Castéras–Gou–Jeanjean [2] deduce ground-wave instability under their stated radial hypotheses. The recent Dinh–Muse manuscript [3], Theorem 1.1 and Remark 1.4, states radial finite-time blowup and leaves the unrestricted nonradial question open. Instability in the full perturbation space is already implied by instability using radial perturbations of the same wave. Thus orbital instability alone should not be presented as resolving the nonradial blowup problem. The distinctive conclusion studied here is a quantitative physical-time average for general near-threshold trajectories, with the precise momentum and spectral assumptions stated in each theorem.

Pausader–Shao [4], Section 6, already relate center motion, momentum, and energy for fixed-scale precompact quartic trajectories with the additional regularity in their proposition. Their rigidity argument rules out the corresponding negative-energy zero-momentum compact trajectory. Our tail argument permits small mass radiation and does not assume kinetic compactness. The remaining moving-tail term is exactly what prevents the present theorem from giving full compact-trajectory rigidity for every bounded solution.

Inui–Komada–Nishimura [5] prove nonradial subground scattering in a range starting at nonlinearity power $1+8/(N-1)$, using a virial transverse to momentum. The mass-critical power $1+8/N$ is outside that range: its transverse scaling exponent is below four. Their argument therefore does not directly settle the critical problem addressed here.

Occupation-sensitive interpolation has a substantial history, including Dolbeault–Felmer–Loss–Paturel [6]. Hong–Kwon–Yoon [7] give fractional operator kinetic estimates and conservation of pairwise component inner products. We do not claim either idea as new. The Gaussian coefficient $\mathcal{B}_r$ above is an explicit refinement retaining the scalar endpoint and the full occupation spectrum; its optimality at fixed higher rank is not asserted. Indeed, for k equal occupations the Gaussian ratio tends to $1/\mathrm{G}(r+2)$ as $k \to \infty$, whereas operator-norm kinetic inequalities can yield mass intervals growing with rank. Neither bound uniformly replaces the other for every purpose.

Frank–Gontier–Lewin [8] prove strict finite-rank binding for the ordinary second-order kinetic problem in their specified range. For eigenvalue moment one and dimension at least three, rank two strictly improves on the rank-one spectral constant. Consequently one cannot simply replace total mass by largest occupation while retaining the scalar optimal kinetic constant. This is an established obstruction from their work; we neither solve a Lieb–Thirring conjecture nor transfer that strict-binding theorem to quartic dispersion without proof.

What remains mathematically open in this manuscript. The mean bounds permit bounded global motion at a sufficiently large kinetic level. They also permit sparse high peaks when no kinetic upper bound is assumed. Applying the same estimate at a later time does not amplify it again, because d and E are conserved. To exclude every bounded negative-energy trajectory one needs a new long-time statement controlling scale evolution or the accumulated motion of the small mass tail. Only after that step could one separate finite-time collapse from infinite-time growth. The modulation identities locate these terms explicitly; they do not remove them. Independently, proving real-sector nondegeneracy in the full ground-wave class would remove the remaining hypothesis from the stronger center theorem. Determining the optimal fixed-occupation constant is another substantive variational problem.

A Scalar compactness and critical normalization

The Fourier transform is unitary, $D = -i\nabla$, and $P(u) = \operatorname{Im} \int \overline{u}\nabla u = \int \xi |\widehat{u}|^2$. The real Hilbert pairing is the real part of the complex L^2 pairing. Throughout the local-power arguments, $N \geq 1$,

$m > 1$, $s = m/2$, $r = m/N$, and $p = 2 + 2r$. A principal symbol a is real, even, m-homogeneous, C^{N+2} off zero, and satisfies $c|\xi|^m \le a(\xi) \le C|\xi|^m$. Set $A(u) = \int a|\widehat{u}|^2$, $M(u) = \|u\|_2^2$, and $E(u) = A(u)/2 - \|u\|_p^p/p$. The sharp critical mass M_c is defined by

$$\|u\|_p^p \le \frac{p}{2}\left(\frac{M(u)}{M_c}\right)^r A(u). \tag{51}$$

The traveling-wave convention $u(t,x) = e^{i\omega t}U(x - vt)$ gives $[a(D) + \omega + iv \cdot \nabla]U - |U|^{2r}U = 0$.

Lemma A.1 (Normalized optimizer compactness)**.** *The inequality (51) is attained. If $M(f_n) \to M_c$, $A(f_n) \to 1$, and $\|f_n\|_p^p \to p/2$, then some translated subsequence converges strongly in H^s to an optimizer. The normalized optimizer set is compact modulo translations and phases.*

Proof. Sharp Sobolev interpolation gives a finite positive best constant, and amplitude and spatial scaling normalize a maximizing sequence as stated. Bounded energy-space sequences that vanish locally in mass converge to zero in L^p: use a bounded-overlap cover by unit balls, local Sobolev interpolation, and $2 < p < 2N/(N - m)$ when $N > m$ (all finite p otherwise). Thus translations give a nonzero weak limit Q and almost-everywhere convergence. Write $M_1 = M(Q) > 0$, $A_1 = A(Q) > 0$, $M_2 = M_c - M_1$, and $A_2 = 1 - A_1$. Quadratic splitting and Brezis–Lieb give

$$M_c^r \le M_1^r A_1 + M_2^r A_2, \qquad A_1 + A_2 = 1.$$

If both masses are positive, the right side is strictly below M_c^r. Consequently $M_1 = M_c$ and then $A_1 = 1$. Weak convergence plus convergence of both quadratic norms gives strong H^s convergence. The same proof applies to a sequence of optimizers. Constant phase is proved by the density-norm lemma below; this compactness argument itself does not need a positivity-preserving kinetic operator. □

B Sharp density geometry and quantitative polarization

This appendix reproduces the analytic inputs used by the new evolution arguments. They are proved here, including the infinite-dimensional compactness step. No radiality, positivity of the kinetic semigroup, or real-sector nondegeneracy is used.

Let $\mathcal{H}$ be any nonzero separable complex Hilbert space. Finite-dimensional $\mathbb{C}^d$ and ℓ^2 are both allowed. Fix $N \ge 1$, $m > 1$, $s = m/2$, and a real-even, positive elliptic, m-homogeneous symbol a, of class C^{N+2} away from zero. For $U \in H^s(\mathbb{R}^N; \mathcal{H})$ put

$$\rho_U = \|U\|_{\mathcal{H}}^2, \quad M(U) = \int \rho_U, \quad A(U) = \int a(\xi)\|\widehat{U}(\xi)\|_{\mathcal{H}}^2\, d\xi.$$

We use either of the following interactions:

$$\begin{aligned} &\kappa = 1 + m/N, && \mathcal{F}(U) = \int \rho_U^\kappa, && V_\rho = \rho^{\kappa-1}; \\ &\kappa = 2, && \mathcal{F}(U) = \mathcal{D}(\rho_U, \rho_U), && V_\rho = K * \rho. \end{aligned}$$

In the second case $1 < m < N$, the real even kernel K is homogeneous of degree $-m$, its interaction form $\mathcal{D}$ is positive definite, and

$$|K(x)| + |x||\nabla K(x)| \le C|x|^{-m}.$$

As before, positivity of the Fourier transform of K almost everywhere suffices for positive definiteness. Pointwise positivity is not needed. The scalar sharp inequality defines M_c by

$$\mathcal{F}(u) \le \kappa\left(\frac{M(u)}{M_c}\right)^{\kappa-1} A(u). \tag{52}$$

Theorem B.1 (Sharp interpolation and dimension-independent compactness)**.** *Inequality* (52) *has the same sharp constant for $\mathcal{H}$-valued functions. Every optimizer has the form*

$$U(x) = \zeta Q(x), \qquad \|\zeta\|_{\mathcal{H}} = 1,$$

where Q is a real scalar optimizer, up to a constant phase. If $M(U_n) \to M_c$, $A(U_n) \to 1$, and $\mathcal{F}(U_n) \to \kappa$, then the sequence is strongly compact in H^s after translations and internal unitaries. This remains true when the internal dimensions vary: embed all spaces in ℓ^2 before applying the unitaries.

B.1 Sharp interpolation and complete polarization

The local interaction is the κ-th power of the L^κ density norm. The Hartree interaction is the square of its positive definite Hilbert density norm. Thus both have the form $\mathcal{F}(U) = \|\rho_U\|_{\mathcal{B}}^\kappa$ for a strictly convex real density space $\mathcal{B}$. In an internal orthonormal basis $U = (u_j)$, the triangle inequality, scalar sharp interpolation, and sequence Hölder give

$$\begin{aligned}
\mathcal{F}(U)^{1/\kappa} &\le \sum_j \mathcal{F}(u_j)^{1/\kappa} \\
&\le \kappa^{1/\kappa} M_c^{-(\kappa-1)/\kappa} \sum_j A(u_j)^{1/\kappa} M(u_j)^{(\kappa-1)/\kappa} \\
&\le \kappa^{1/\kappa} M_c^{-(\kappa-1)/\kappa} A(U)^{1/\kappa} M(U)^{(\kappa-1)/\kappa}.
\end{aligned}$$

Finite partial sums converge in the density norm because the same bound applied to their tails tends to zero. A scalar maximizing sequence in one component proves sharpness.

Split the components further into real scalar functions f_j. Equality forces every active f_j to be a scalar optimizer, all f_j^2 to be proportional, and all kinetic-to-mass ratios to agree. The same conclusions hold after any real orthogonal rotation of a pair f_i, f_j. If f_j/f_i took both signs of a nonzero constant, the squared sum and difference could not be proportional; when the constant has magnitude one their nonzero supports would be disjoint. Thus each f_j is one real constant multiple of a fixed scalar function. This proves complete polarization, including for countably many components.

B.2 Compactness modulo internal unitaries

Ordinary local compactness does not hold directly in an infinite internal dimension: a fixed spatial shape can move through mutually orthogonal internal vectors. The following argument removes precisely that loss of compactness.

Lemma B.2 (Internal-unitary local compactness)**.** *Let $X = H^s(\mathbb{R}^N; \mathbb{C})$ with $s > 0$. From every bounded sequence in $X \widehat{\otimes} \ell^2$, one can extract a subsequence and internal unitaries W_n such that $W_n U_n$ converges weakly in $X \widehat{\otimes} \ell^2$ and strongly in $L^2_{\mathrm{loc}}(\mathbb{R}^N; \ell^2)$.*

Proof. Fix an orthonormal basis (ϕ_k) of the scalar Hilbert space X. Write its internal coefficient vectors as $c_{nk} = \langle \phi_k, U_n \rangle_X \in \ell^2$. Gram–Schmidt the first n vectors and extend the resulting finite orthonormal list to an internal orthonormal basis. A unitary W_n can therefore be chosen so that

$$W_n c_{nk} \in \operatorname{span}\{e_1, \dots, e_k\}, \qquad k \le n.$$

For each fixed k, these vectors have a convergent subsequence in a fixed finite-dimensional space. Diagonal extraction gives strong convergence of every coefficient vector. Weak compactness gives a weak limit of $W_n U_n$, with exactly those coefficients.

For $\chi \in C_c^\infty$, the scalar multiplication map $K_\chi : X \to L^2$, $f \mapsto \chi f$, is compact. If P_L is the scalar projection onto the first L basis vectors, then

$$\|K_\chi(I - P_L)\|_{X \to L^2} \longrightarrow 0.$$

Tensoring with the identity on ℓ^2 preserves this norm. The transformed $P_L U_n$ converge strongly because there are only L coefficient vectors; after multiplication by χ, the tails are uniformly small. This proves local strong convergence. □

Compactness assertion of Theorem B.1. The localized Sobolev inequality has constants independent of internal dimension: use the Hilbert-valued Bessel-potential representation and its norm triangle inequality, then spatial localization and the sum of component quadratic norms. Thus local mass vanishing for an H^s-bounded sequence implies vanishing in $L^p(\mathcal{H})$ for $2 < p < 2N/(N-2s)$, or every finite $p > 2$ if $2s \geq N$. In the local case take $p = 2\kappa$; in the Hartree case use $p = 4N/(2N-m)$ and Hardy–Littlewood–Sobolev. Hence $\mathcal{F}(U_n) \to \kappa > 0$ excludes local mass vanishing.

Translate a nonzero amount of local mass into a fixed ball and use Lemma B.2. We obtain a nonzero weak limit U, strong local L^2 convergence, and by interpolation strong local convergence in the interaction exponent. Put $R_n = U_n - U$. The quadratic quantities split. For local powers the Brezis–Lieb argument for $\|U\|_{\mathcal{H}}^{2\kappa}$ gives

$$\mathcal{F}(U_n) = \mathcal{F}(U) + \mathcal{F}(R_n) + o(1).$$

For Hartree, the term $\mathrm{Re}\langle U, R_n\rangle$ tends to zero in $L^{p/2}$ by local convergence and the small L^p tail of U. Also $\mathcal{D}(\rho_U, \rho_{R_n}) \to 0$: approximate U in $L^p(\mathcal{H})$ by a compactly supported smooth finite-dimensional field. Its potential is bounded and decays at infinity, while R_n tends locally to zero in L^2. HLS controls the approximation error. This proves the same splitting for Hartree.

Let $M_1 > 0, M_2$ and A_1, A_2 be the limiting contributions. Then $M_1 + M_2 = M_c$, $A_1 + A_2 = 1$, and sharp interpolation yields

$$M_c^{\kappa-1} \leq M_1^{\kappa-1} A_1 + M_2^{\kappa-1} A_2.$$

Both masses cannot be positive, since each would be strictly below M_c. Thus $M_2 = 0$, and the inequality then forces $A_2 = 0$. The convergence is strong in H^s. The limit is an optimizer, already proved to be completely polarized. □

B.3 Phase coercivity and the internal symmetry kernel

For completeness, the scalar phase fact used below follows directly from the density norm. Normalize a real optimizer by $M(Q) = M_c$, $A(Q) = 1$, and put $\omega_* = (\kappa - 1)/M_c$. The form

$$L_Q = a(D) + \omega_* - V_{Q^2}$$

is nonnegative: differentiate the sharp inequality at $Q + i\sqrt{t}\, h$. If h belongs to its kernel, write $x = A(h)$ and $y = M(h)/M_c$. The supporting-functional inequality for the density norm, sharp interpolation for h, and weighted arithmetic–geometric mean give

$$x + (\kappa - 1)y \leq \kappa x^{1/\kappa} y^{(\kappa-1)/\kappa} \leq x + (\kappa - 1)y.$$

Equality implies $x = y$ and $h^2 = yQ^2$ by strict convexity. The kernel of a nonnegative form is linear; apply the same conclusion to $h + Q$ to get $h = cQ$. The potential is compact as a form $H^s \to H^{-s}$, by local Rellich and Hölder in the local case, or the compact-potential approximation and HLS in the Hartree case. Consequently

$$\langle h, L_Q h\rangle \geq c_Q \|h\|_{H^s}^2, \qquad \langle h, Q\rangle_{L^2} = 0. \tag{53}$$

This holds for complex h by real–imaginary splitting, and it can be summed over arbitrarily many components with the same constant.

At Qe_1, write a perturbation as $(h+ik)e_1 + Z$ with $Z \perp e_1$. The Hessian of $E + \omega_* M/2$ is

$$\langle h, L_+ h\rangle + \langle k, L_Q k\rangle + \sum_{j\geq 2}\big(\langle \operatorname{Re} Z_j, L_Q \operatorname{Re} Z_j\rangle + \langle \operatorname{Im} Z_j, L_Q \operatorname{Im} Z_j\rangle\big).$$

Its kernel in the phase and internal transverse sector is exactly

$$\{Q\eta :\ \eta \in \mathcal{H}, \operatorname{Re}\langle \eta, e_1\rangle = 0\}.$$

For $\mathcal{H} = \mathbb{C}^d$, this is the $(2d-1)$-dimensional real tangent space to the internal unitary orbit. Any additional full Hessian kernel lies in the scalar real sector L_+; no assumption on that sector is made.

B.4 Quantitative polarization without real-sector nondegeneracy

The preceding compactness and phase coercivity also give a quantitative statement for arbitrary fields, not just stationary or traveling ones. Define the scale-invariant interpolation deficit

$$\mathfrak{d}(U) = 1 - \frac{\mathcal{F}(U)}{\kappa(M(U)/M_c)^{\kappa-1}A(U)}$$

and the squared distance to a completely polarized field by

$$\mathfrak{p}(U) = \inf_{\substack{\|\zeta\|_{\mathcal{H}}=1 \\ q\in H^s}} \left\{ \frac{M(U-\zeta q)}{M(U)} + \frac{A(U-\zeta q)}{A(U)} \right\}.$$

Both are defined for every nonzero $U \in H^s(\mathcal{H})$.

Theorem B.3 (Dimension-independent quantitative polarization)**.** *There is $C < \infty$, depending only on the fixed scalar dispersion and interaction, such that*

$$\mathfrak{p}(U) \leq C\mathfrak{d}(U) \tag{54}$$

for every nonzero Hilbert-valued field U and every internal dimension. No nondegeneracy or uniqueness of the scalar real-sector ground state is required.

Proof. Amplitude and spatial scaling preserve both quantities, so impose $M(U) = M_c$, $A(U) = 1$. Write $\|U\|_X^2 = M(U)/M_c + A(U)$. Since $\mathfrak{p}(U) \leq 2$, failure of a uniform constant would produce U_n with $\mathfrak{d}(U_n) \to 0$ and $\mathfrak{d}(U_n)/\mathfrak{p}(U_n) \to 0$. After the symmetries of Theorem B.1, $U_n \to Qe_1$ strongly in H^s. Align $\int QU_n$ with the positive real first internal direction and write $U_n = (u_n, Z_n)$. Each component of Z_n is complex L^2-orthogonal to Q, and $u_n \to Q$, $Z_n \to 0$ in H^s.

The interaction expansion in these transverse directions is

$$\mathcal{F}(u_n, Z_n) = \mathcal{F}(u_n) + \kappa\langle V_{|u_n|^2}, \|Z_n\|^2\rangle + o(\|Z_n\|_X^2). \tag{55}$$

For local powers, set $a = |u_n|^2$ and $b = \|Z_n\|^2$ pointwise. When $1 < \kappa \leq 2$, the nonnegative Taylor remainder of $(a+b)^\kappa$ after its linear term in b is at most $C_\kappa b^\kappa$. When $\kappa > 2$, it is at most $C_\kappa(a^{\kappa-2}b^2 + b^\kappa)$. Hölder and the dimension-independent $L^{2\kappa}$ Sobolev bound make these respectively $O(\|Z_n\|_X^{2\kappa})$ and $O(\|Z_n\|_X^4 + \|Z_n\|_X^{2\kappa})$, since u_n stays bounded. Both are $o(\|Z_n\|_X^2)$. For Hartree the remainder is exactly $\mathcal{D}(\|Z_n\|^2, \|Z_n\|^2) = O(\|Z_n\|_X^4)$.

Put $m_n = M(Z_n)$ and $a_n = A(Z_n)$. Scalar sharp interpolation gives

$$\mathcal{F}(u_n) \leq \kappa(1 - m_n/M_c)^{\kappa-1}(1 - a_n).$$

Combining its expansion with (55) and potential form convergence yields

$$\begin{aligned}\kappa - \mathcal{F}(U_n) &\geq \kappa\left[a_n + \frac{\kappa-1}{M_c}m_n - \langle V_{Q^2}, \|Z_n\|^2\rangle\right] - o(\|Z_n\|_X^2)\\ &= \kappa\sum_j \langle Z_{nj}, L_Q Z_{nj}\rangle - o(\|Z_n\|_X^2)\\ &\geq c\|Z_n\|_X^2.\end{aligned}$$

The constant is independent of component number. Taking the candidate $e_1 u_n$ in the distance gives $\mathfrak{p}(U_n) \leq \|Z_n\|_X^2$. As $\mathfrak{d}(U_n) = (\kappa - \mathcal{F}(U_n))/\kappa$, this contradicts the assumed ratio and proves the theorem. □

The theorem controls internal polarization rather than the full scalar shape. It therefore remains valid even when scalar optimizer uniqueness or real-sector quantitative stability is unavailable.

B.5 The real version and scalar constant phase

All preceding density inequalities, compactness, and quantitative polarization statements hold over a separable real internal Hilbert space, with orthogonal maps replacing unitaries and real scalar q in the definition of $\mathfrak{p}$. Here are the points that matter for the scalar evolution theorem. The real scalar sharp constant equals the complex one: apply the density triangle inequality and sequence Hölder to $(\operatorname{Re} u, \operatorname{Im} u)$, starting with the real scalar constant, while the reverse comparison is immediate. Equality forces a constant real internal direction by the pair-rotation argument above. The internal compactness proof uses real Gram–Schmidt without any other change. Finally rotate $\int QU_n$ into the first real direction. Every transverse real component is orthogonal to Q, so exactly the displayed L_Q coercivity and transverse expansion apply. This proves the real quantitative assertion, not just its complex analogue.

In particular, for a scalar complex u and either allowed interaction, there exists a phase θ such that $u = e^{i\theta}(X + iY)$, with X, Y real, and

$$\frac{M(Y)}{M(u)} + \frac{A(Y)}{A(u)} \leq C\mathfrak{d}(u).$$

This holds for every local exponent $r > 0$, and for the strictly positive-definite Hartree density form. For negative energy and $d = (M/M_c)^{\kappa-1} - 1 > 0$, one has $0 \leq \mathfrak{d}(u) < d/(1+d)$, providing exactly the input used in Theorem 3.1. No phase is differentiated in time. In the local range $r \geq 1$, Theorem 1.2 gives an explicit constant in place of this compactness argument.

C Zero momentum and instantaneous quartic transport

The distinction between momentum and quartic mass transport is exact. Choose a real sharp wave Q and a real smooth test h with

$$\langle h, Q\rangle = 0, \qquad \langle h, \partial_k Q\rangle = 0 \ (1 \leq k \leq N), \qquad \langle h, \partial_j \Delta Q\rangle \neq 0$$

for some j. Such a test exists. Otherwise $\partial_j \Delta Q \in \operatorname{span}\{Q, \partial_1 Q, \ldots, \partial_N Q\}$ as distributions. Pairing with Q removes its Q component, and Fourier transformation would force $\widehat{Q}$ onto the zero set of a nonzero polynomial, contradicting $Q \neq 0$ in L^2. For fixed sufficiently large $a > 0$, set

$$u_\varepsilon = (1 + a\varepsilon^2)Q + i\varepsilon h.$$

Then $P(u_\varepsilon) = 0$ exactly, and

$$d_\varepsilon = \frac{4}{N}\left(2a + \frac{M(h)}{M_c}\right)\varepsilon^2 + O(\varepsilon^4),$$
$$E(u_\varepsilon) = \varepsilon^2\left[\frac{A(h) - \int |Q|^{8/N}h^2}{2} - a\omega M_c\right] + o(\varepsilon^2) < 0,$$

where $\Delta^2 Q + \omega Q - |Q|^{8/N}Q = 0$. Nevertheless its quartic mass current is

$$4\int \xi_j |\xi|^2 |\widehat{u}_\varepsilon(\xi)|^2 \, d\xi = 8\varepsilon(1 + a\varepsilon^2)\langle h, \partial_j \Delta Q\rangle.$$

Thus this current has order $\sqrt{d_\varepsilon}$ even at exactly zero momentum and negative energy. With a finite first moment it is the instantaneous derivative of that moment; without one it remains the finite Fourier current underlying localized mass transport.

The improved center in Section 4 is a modulation coordinate chosen by dual orthogonality, not the ordinary first-moment center. Its quadratic drift is therefore consistent with the example. The example rules out the proposed instantaneous current improvement; it does not rule out further cancellation along the nonlinear evolution.

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